

## NETWORKS AND PROTOCOLS

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# Exercises Sheet 1

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### Exercise 1

An image has a size of 4080x1830 pixels with true color, which means that 3 Bytes per pixel are used for the color information.

How long does it take to transmit the uncompressed image via a:

- (a) 56 kbps Modem connection?
- (b) 64 kbps ISDN connection?
- (c) 16 Mbps DSL connection?
- (d) 100 Mbps Ethernet connection?
- (e) 1 Gbps Ethernet connection?

Assume the image is compressed with a compression algorithm that reduces the image size by 85%. How long does it take to transmit the image via a :

- (a) 1 Mbps DSL connection?
- (b) 10 Mbps Ethernet connection?
- (c) 10 Gbps Optical connection?

A MP3 file with a size of  $30 \cdot 10^6$  bits must be transferred from terminal device A to terminal device B. The signal propagation speed is 200,000 km/s. A and B are directly connected by a link with a length of 5,000 km. The file is transferred as a single message that has a size of  $30 \cdot 10^6$  bits. No network protocol headers or trailers exist.

Calculate the transfer time (latency) of the file, when the data rate of the computer network between both terminal devices is:

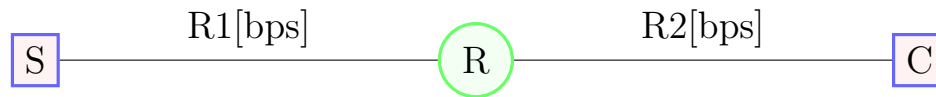
- (a) 56 kbps Modem connection?
- (b) 64 kbps ISDN connection?
- (c) 100 Mbps Ethernet connection?
- (d) 1 Gbps Ethernet connection?

Calculate for each one of the above alternatives what the volume of the network connection is. What is the maximum number of bits that can reside inside the line between the sender and receiver?

## Exercise 2

Assume there is one router and two links between the file server (S) and client (C), as shown in the figure below. The first link has transmission rate  $R_1$  and the second link has transmission rate  $R_2$ . Assume the file gets broken into three packets, each of size  $L$ . Ignore all propagation and processing delays. Answer the following three questions:

- (a) How long does it take from when the server starts sending the file until the client has received the whole file if  $R_1 \leq R_2$ ?
- (b) How long does it take from when the server starts sending the file until the client has received the whole file if  $R_1 > R_2$ ?
- (c) In case (b), how long does the second packet spend in the router's queue?



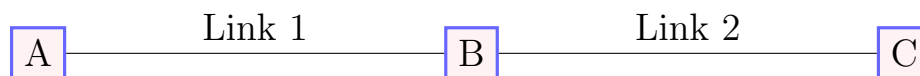
## Exercise 3

Consider a single link with bandwidth  $B$  and propagation delay  $L$ . It takes 1 ms for an entire 500 bytes packet to arrive at the other end of the link (that is, it takes 1 ms from the time the first bit starts being transmitted until the last bit arrives at the other end of the link). It takes 2 ms for an entire 1500 bytes packet to arrive at the other end of the link.

- (a) What is the bandwidth  $B$  of the link? (in Mbps)
- (b) What is the propagation delay  $L$  of the link? (in ms)

## Exercise 4

Consider the diagram below. Link 1 has length  $L_1$  m (where m stands for meters) and allows packets to be propagated at speed  $S_1$  (m/s), while Link 2 has length  $L_2$  m but it only allows packets to be propagated at speed  $S_2$  (m/s) (because the two links are made of different materials). Link 1 has transmission rate  $T_1$  (bit/s) and Link 2 has transmission rate  $T_2$  (bit/s).



Assuming nodes can send and receive bits at full rate and ignoring processing delay, consider the following scenarios:

- (a) How long would it take to send a packet of 500 Bytes from Node A to Node B given  $T_1 = 10000$  bit/s,  $L_1 = 100000$  m, and  $S_1 = 2.5 \cdot 10^8$  m/s?

- (b) Compute RTT (round trip time) for a packet of  $B$  Bytes sent from Node A to Node C (packet gets transmitted back from Node C immediately after Node C receives it).
- (c) At time 0, Node A sends packet  $P_1$  with  $D_1$  Bytes and then it sends another packet  $P_2$  with  $D_2$  Bytes immediately after it pushes all bits of  $P_1$  onto Link 1. When will Node C receive the last bit of  $P_2$ ?

### Exercise 5

Unlike store-and-forward, cut-through switching or forwarding allows a node inside a network to start forwarding a packet before it has been received in its entirety. Consider a path in a network that connects a host A to a host B via  $n$  links (1.. $n$ ) through  $n-1$  intermediate routers, each employing cut-through forwarding.



Ignore processing delay and assume that there is no queuing delay. Assume that all links have the same length  $d$  and that the speed of light in the link medium is  $s$ . Also, assume that each link has bandwidth  $R$  and that packets of length  $L$  bits are sent through the network. Each switch can start forwarding the packet after  $L_c$  bits have been received.

- (a) What is the total latency for a single packet (counted from first bit sent at A until last bit received at B)?
- (b) How much did cut-through switching reduce the latency for sending a single packet end-to-end, compared to store-and-forward?